

Maroua Arbouni

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SUMMARY

AI & Digital Transformation Engineering Student equipped with strong analytical skills and expertise ranging from web development (Laravel, PHP) to enterprise architecture and artificial intelligence integration. Passionate about designing innovative digital platforms, optimizing business processes, and creating high-value technological solutions.

EDUCATION

National School of Applied Sciences (ENSA)

Engineering Cycle: AI & Digital Transformation

Al Hoceima, Morocco

2025 – Expected 2028

National School of Applied Sciences (ENSA)

Integrated Preparatory Classes

Al Hoceima, Morocco

2023 – 2025

PROJECTS

PFE Assignment & Management System | *Laravel, MySQL, Bootstrap, HTML, CSS*

- Automated student-supervisor assignments under strict equitable constraints by engineering an application that parses comprehensive Excel uploads of students across 3 majors and available professors.
- Streamlined defense scheduling and eliminated overlaps by developing an automated planning module that dynamically coordinates supervisors, juries, students, rooms, and timeslots.

Building Permit Management Platform | *Laravel, MySQL, Python (NLP), Taiga*

- Accomplished full digitalization of building permit lifecycles by designing a comprehensive web platform featuring online submission, multi-actor validation workflows, and digital archiving.
- Optimized municipal workload distribution and reduced processing time by integrating Python-based NLP scripts to automatically extract applicant data and verify document compliance.

Universal Data Analyzer | *Python, Pandas, Streamlit, PySide6, SQLite*

- Automated cross-file data consolidation and reporting for rapid insights by engineering a Python application that processes multiple Excel uploads to generate unified summary files.
- Enhanced data visualization and analytical capabilities by dynamically generating comprehensive charts and consolidated information within the final exported reports.

Web Platform "CozyWatch" | *Laravel, PHP, Bootstrap, HTML, CSS* | [GitHub](#)

- Developed a movie management platform using Laravel MVC, featuring full CRUD operations and a robust database schema.

Pharmacy Management System (MediControl) | *C, Linked Lists, Advanced Algorithms* | [GitHub](#)

- Developed an application in C utilizing linked lists to handle pharmacy stock management, order tracking, and centralized low-stock notifications, etc.

Boarding School Services Digitalization | *ArchiMate, Zachman Framework, SWOT, PESTEL, Porter*

- Designed the enterprise architecture (AS-IS to TO-BE) for a boarding school platform covering student accommodation, secure payments, maintenance ticketing, and BI reporting, using ArchiMate and Zachman.

Custom Automata Compiler | *C, Flex, Bison, Graphviz*

- Created "Language A", a custom language compiled with Flex and Bison, designed to generate and simulate automata with semantic error detection and Graphviz exports.

TECHNICAL SKILLS

Languages: PHP, Python, Java (OOP), C, HTML, CSS

Frameworks & Systems: Laravel, Bootstrap, Linux, MySQL, Information Systems & Databases

Architecture & Modeling: ArchiMate, SWOT, PESTEL, Porter, Language Theory

Tools & Industry 4.0: Git, Taiga, openPLC (Digital Industry)